

Weeks 7 & 8 Assignment - Lottery Numbers

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DSC640-T301 Data Presentation & Visualizat (2263-1)

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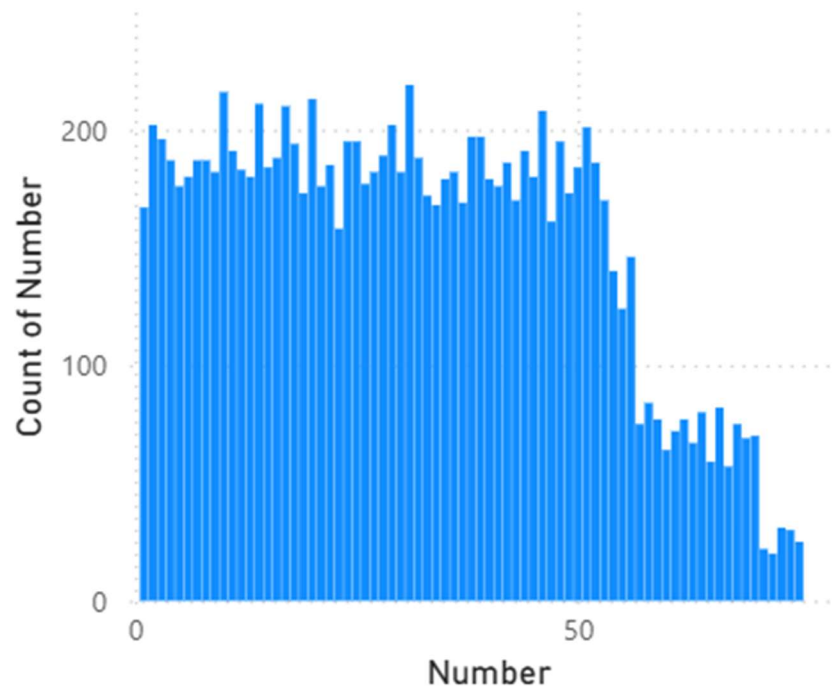
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Visuals

Figure 1. Mega Millions Number Frequency (Histogram)

Winning numbers appear evenly distributed across the full range.

Mega Millions: Number Frequency (Histogram)

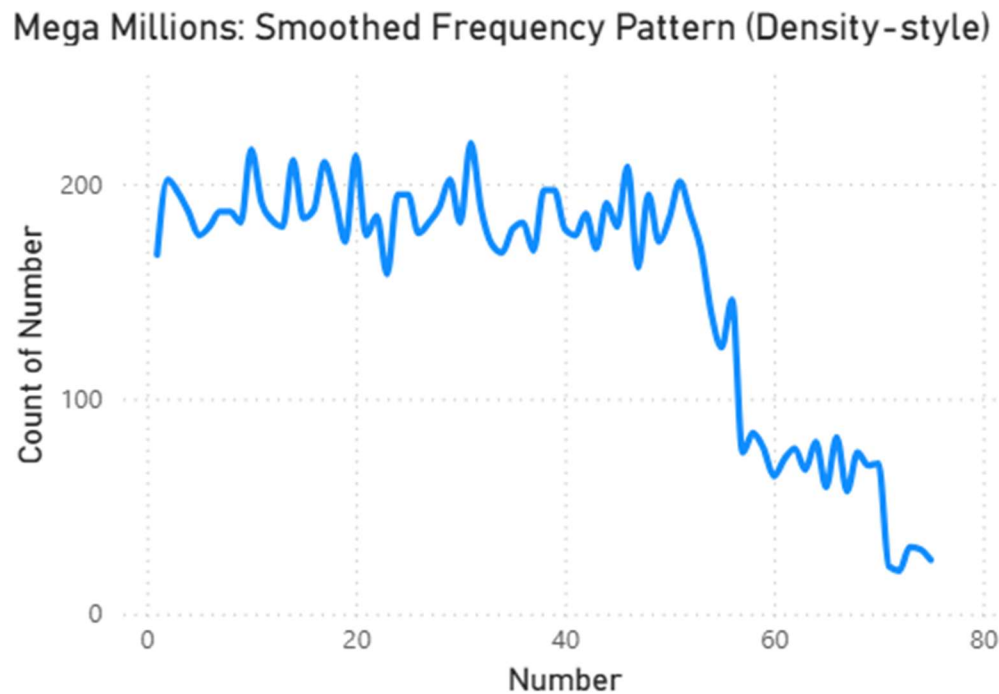


Interpretation:

Winning numbers occur at relatively similar frequencies over time, with no evidence that certain numbers appear more often in a statistically meaningful way. This suggests that perceived “hot” numbers are a normal artifact of randomness rather than an indicator of future outcomes.

Figure 2. Mega Millions Smoothed Frequency Pattern (Density-style)

Short-term fluctuations smooth out over long time horizons.

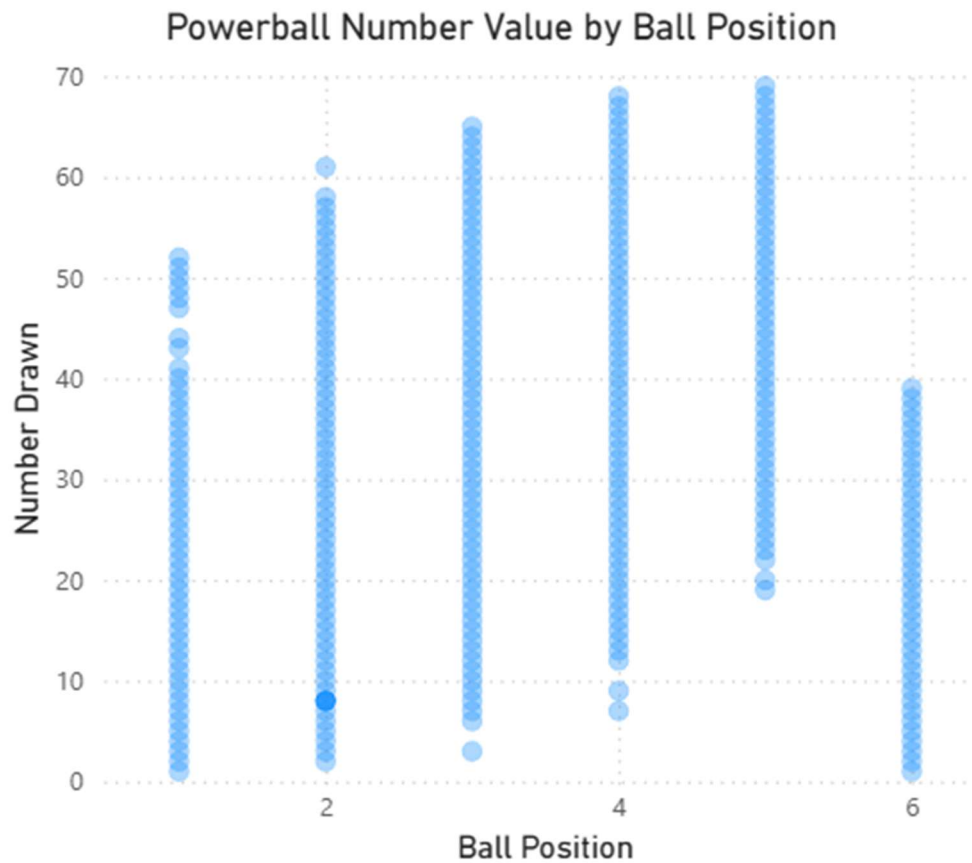


Interpretation:

While the smoothed curve shows minor peaks and dips, these variations do not persist consistently across the number range. This shows that apparent clustering emerges naturally from randomness rather than from any stable or predictable structure.

Figure 3. Powerball Number Value by Ball Position

Ball position does not influence the value of the number drawn.

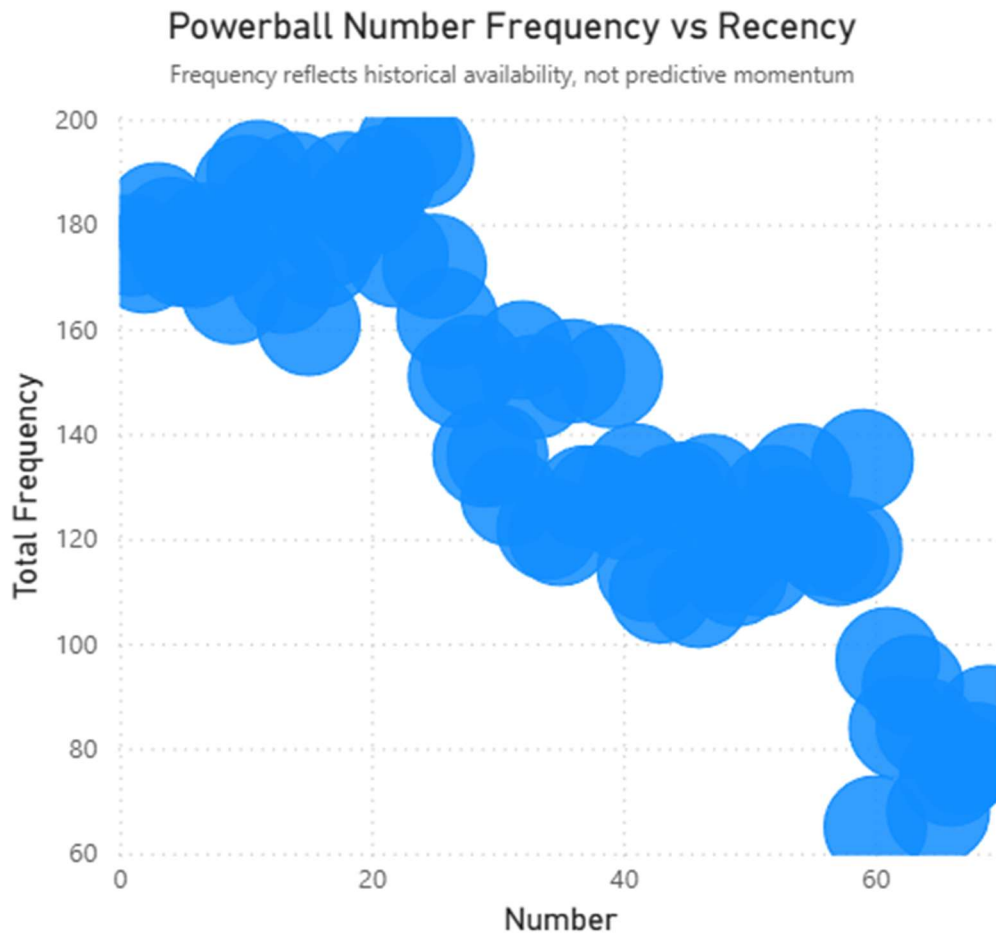


Interpretation:

The scatterplot shows that each ball position contains a wide and overlapping range of values, with no visible trend or separation by position. This indicates that draw order has no relationship to the magnitude of the number selected, reinforcing that perceived positional patterns do not create predictive advantage.

Figure 4. Powerball Number Frequency vs Recency

Frequency reflects historical availability, not predictive momentum.

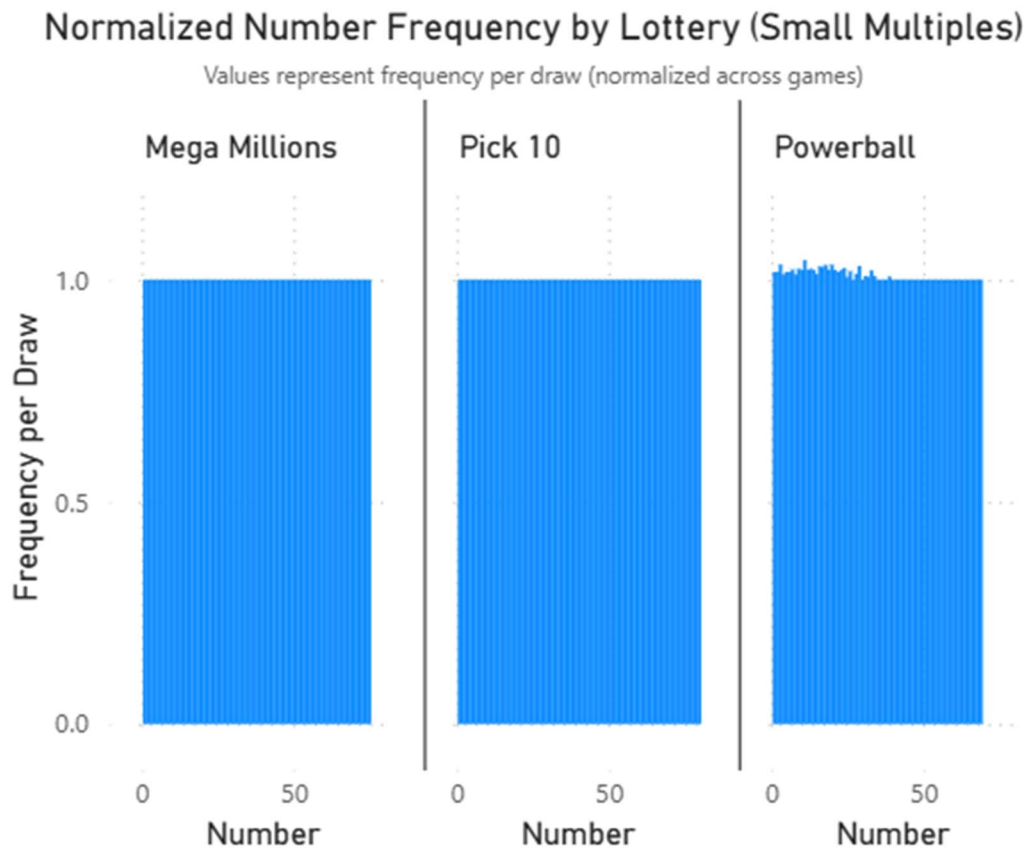


Interpretation:

Numbers with higher historical frequency are not consistently more recent, and recently drawn numbers are not systematically more frequent. This disconnect shows that neither frequency nor recency provides meaningful momentum for predicting future Powerball outcomes.

Figure 5. Normalized Number Frequency by Lottery (Small Multiples)

Values represent frequency per draw, normalized across games.

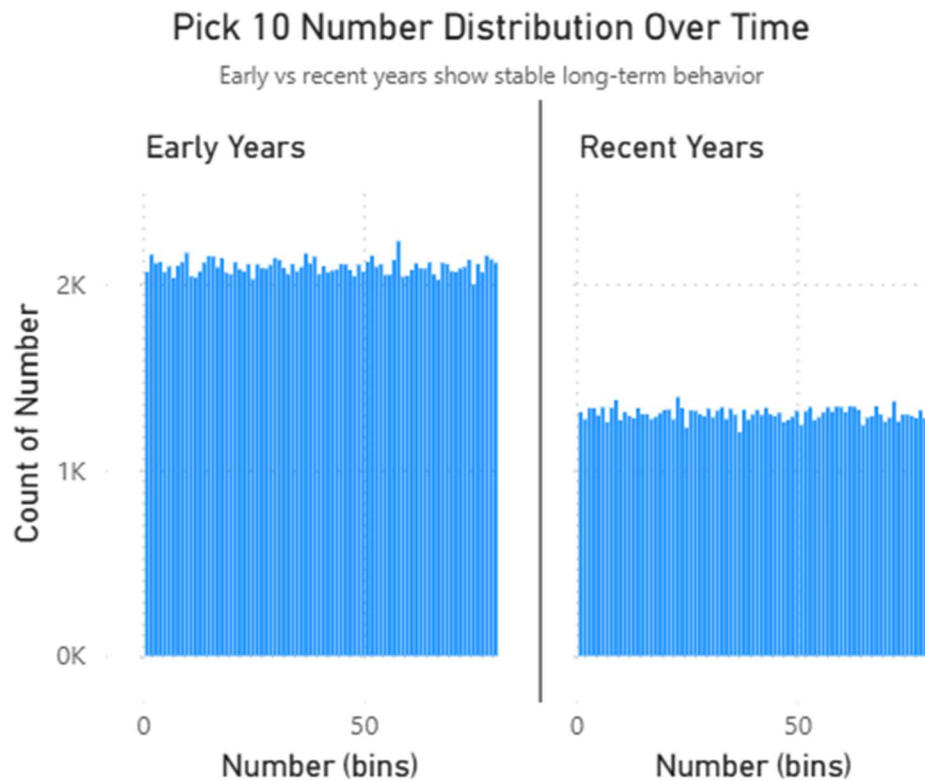


Interpretation:

When frequencies are normalized by draw structure, all three lotteries display nearly flat distributions with only minor natural variation. Despite differences in game design and number ranges, each lottery exhibits the same underlying random behavior, reinforcing that structural differences between games do not create predictive advantage.

Figure 6. Pick 10 Number Distribution Over Time

Early vs recent years show stable long-term behavior.



Interpretation:

Although short-term differences appear between early and recent periods, the overall distribution shape remains consistent over time. This indicates that short-term variation fades over time and that Pick 10 outcomes remain governed by long-term randomness.

Summary

Audience

The intended audience for this analysis is the general public and casual lottery players. This audience is expected to have low to moderate data literacy. While they are generally familiar with basic charts such as histograms and scatterplots, they may not have a strong understanding of probability or statistical theory. Because of this, the visuals prioritize clarity, intuitive layouts, and minimal jargon. Titles and subtitles are used to guide interpretation and reduce the risk of misreading patterns as meaningful signals.

Purpose

The purpose of this presentation is to challenge the belief that lottery numbers exhibit exploitable patterns that can improve a player's chances of winning. Many lottery participants rely on strategies such as choosing frequently drawn numbers, avoiding recently drawn numbers, or favoring specific draw positions. This analysis visually demonstrates that these strategies are unsupported by historical data and that apparent patterns are natural outcomes of randomness rather than indicators of predictability. Each visual addresses a distinct question about frequency, position, recency, structure, or time to ensure that conclusions are drawn from multiple independent analytical perspectives.

Call to Action

Play the lottery for entertainment rather than expectation. Avoid selecting numbers based on perceived patterns, frequency, recency, or draw position, as these do not improve the probability of winning.

Medium

The analysis is presented using a Power BI dashboard. This medium allows multiple visuals to be viewed together, supporting comparison and reinforcing the narrative across datasets. Power BI was selected because it enables consistent formatting, supports a wide range of chart types, and allows complex datasets to be transformed into accessible visual stories for a general audience. Although Power BI supports interactive exploration, the visuals are intentionally presented as a static narrative to guide interpretation and reduce the risk of misreading random variation as meaningful patterns.

Design Considerations

Gestalt principles guided the visual design choices throughout the dashboard. Similar color schemes and consistent chart layouts reinforce relationships between visuals, while alignment and spacing help guide the viewer's attention across the presentation. Small multiples are used to compare different lottery systems side by side without overlapping data, making differences and similarities easier to interpret. Titles and subtitles are intentionally descriptive to prevent viewers from inferring predictive meaning where none exists. Across visuals, observed variation remains within expected random fluctuation rather than indicating statistically meaningful deviation from uniform behavior.

Ethical Considerations

All data used in this analysis was obtained from publicly available sources provided by data.ny.gov. No personal or sensitive information is included. The analysis is intentionally framed to avoid implying predictive power or encouraging harmful

gambling behavior. Normalization and clear labeling were applied to prevent misleading comparisons across lottery systems with different structures. Any visible variation in the data is explicitly framed as expected sampling fluctuation rather than evidence of systematic bias or strategic opportunity. This analysis assumes that the core draw mechanics for each lottery remained consistent over time and that publicly reported results accurately reflect official outcomes.

Conclusion

Together, these visuals show that perceived patterns in lottery numbers are expected outcomes of randomness rather than indicators of future success. In a truly random system, long-run frequencies are expected to approximate a uniform distribution, and the observed results align with that expectation. This consistency holds across frequency distributions, positional analysis, structural comparisons, and time-based evaluation. While short-term variation can appear compelling, long-term behavior remains stable across games and time. The safest conclusion is to treat lottery participation as entertainment, not strategy.